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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

BACKEND PROGRAMMING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	Give examples for REPL commands.	L1	CO1	**02**
	b.	Explain the role of the V8 engine in Node.js.	L1	CO2	**02**
	c.	List the key http methods.	L2	CO3	**02**
	d.	What are the advantages of Reactjs?	L1	CO4	**02**
	e.	List the types of NoSQL databases.	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Discuss the features of node.js.	L6	CO1	**05**
		OR			
	b.	Write a Program using JavaScript to find sum of N natural numbers	L2	CO1	**05**
3.	**a.**	Explain the role of Node Package Manager for JavaScript development.	L2	CO2	**05**
		OR			
	b.	How can you manually create an Express.js application without using a generator? Outline the steps.	L1	CO2	**05**
4.	**a.**	Illustrate to steps to install express and develop sample express.js application.	L2	CO3	**05**
		OR			
	b.	Explain how to configure an Express.js application to serve static files such as images or CSS.	L2	CO3	**05**
5.	**a.**	Describe the working of Restful API.	L2	CO4	**05**
		OR			
	b.	Write a simple React class component and explain how it differs from a functional component.	L2	CO4	**05**
6.	**a.**	Discuss in detail about Various Component Types of React.js with example.	L2	CO4	**05**
		OR			
	b.	Explain the Mounting Phase of a component in react.js	L2	CO4	**05**
7.	**a.**	Explain CRUD operations of Mongo dB with example.	L2	CO5	**05**
		OR			
	b.	What are the steps to create a new database in MongoDB using the command line?	L2	CO5	**05**

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		PART-C	6 X 10=60		
8.	a.	Discuss the importance of mongoose in node.js development.	L2	CO1	10
		OR			
	b.	Analyze how front-end and back-end separation improves the development process and code maintainability.	L4	CO1	10
9.	a.	Develop a node.js application with additional routes and logging a middleware	L4	CO2	10
		OR			
	b.	Explain the role of the require function in a Node.js application.	L2	CO2	10
10.	a.	Develop a code for app.js with configurations	L6	CO3	10
		OR			
	b.	Analyze how environment configuration affects an Express.js application. Discuss best practices for managing configuration in different environments (development, staging, production).	L4	CO3	10
11.	a.	Explain different RESTful route patterns to follow when creating an application.	L2	CO3	10
		OR			
	b.	Build an Express.js API with RESTful routes that allows users to register and manage accounts (sign-up, update profile, and delete account). Explain your design decisions and the routing structure.	L6	CO3	10
12.	a.	Explain the important features of react.js	L2	CO4	10
		OR			
	b.	Explain the steps for manually setting up a React application without using create-react-app including all required dependencies.	L2	CO4	10
13.	a.	Design and implement a REST API using Express.js and MongoDB for user management. The API should allow users to register, login, update their profiles, and manage their data.	L4	CO5	10
		OR			
	b.	Develop code to join two collections in MongoDB using node.js	L6	CO5	10